CS223 Computer Architecture & Organization

Instruction Execution Principles

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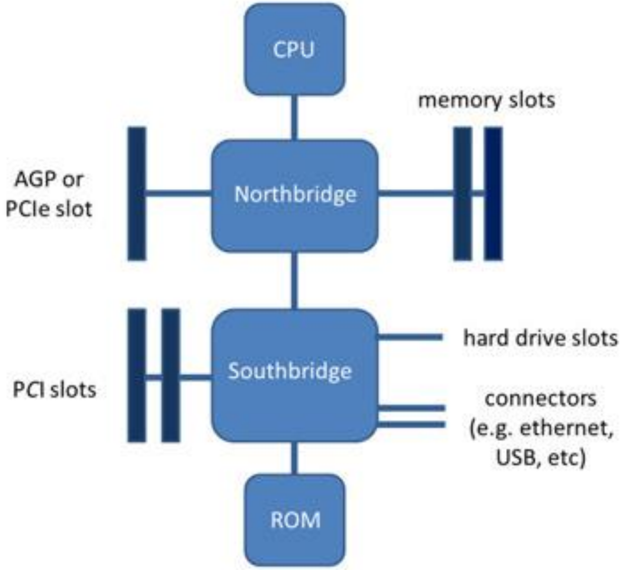
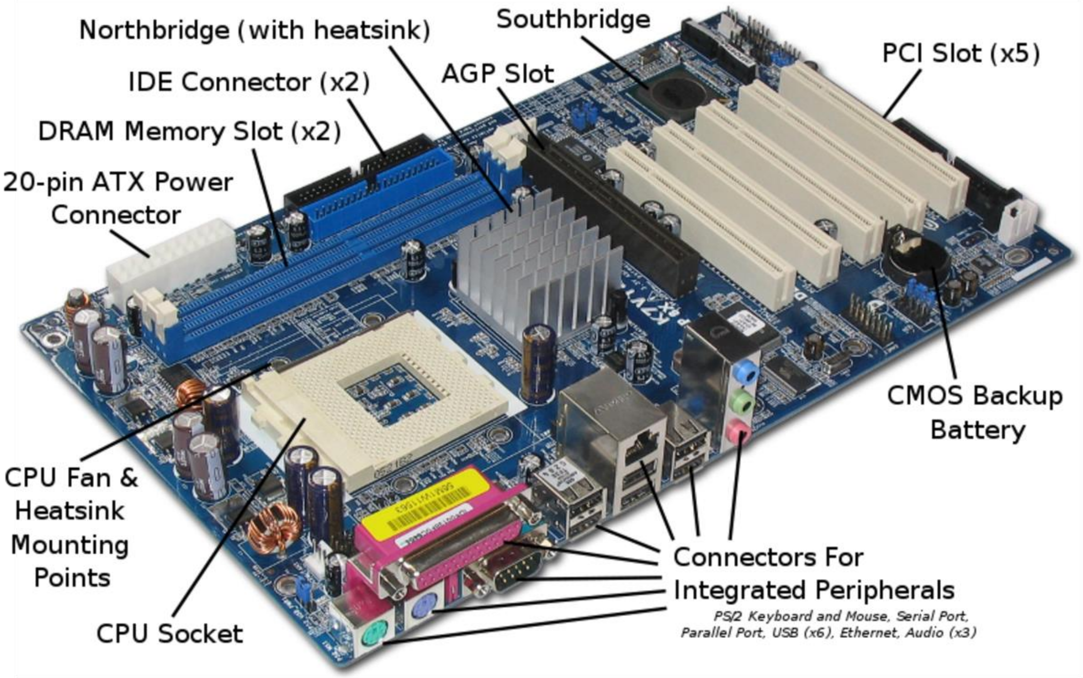
Computer Organization vs Architecture

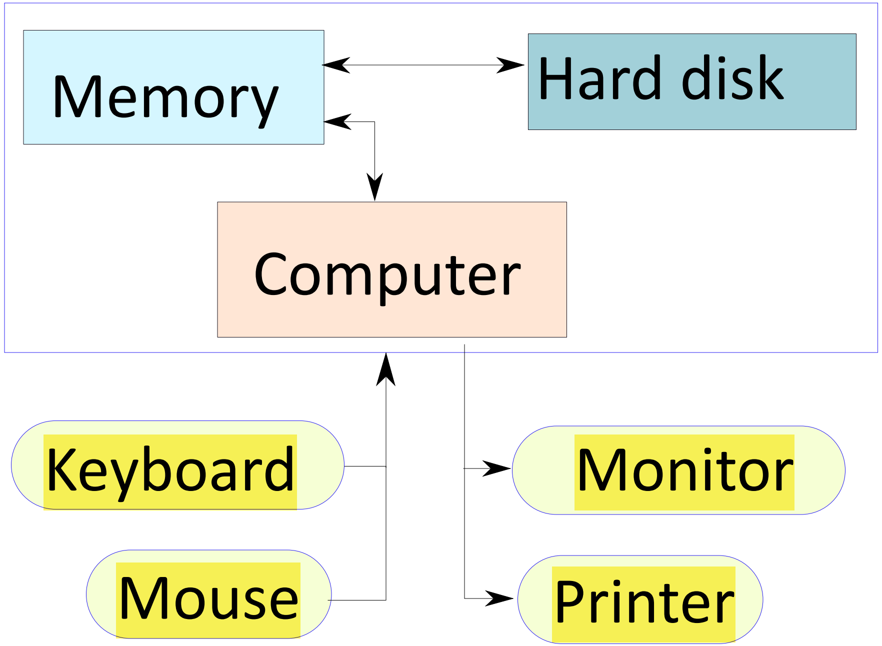
- **Computer Organization**
  - The actual implementation of a computer in hardware
  - Transparent to programmer
  - How to do? (Implementation, Interaction between units)
- **Computer Architecture**
  - The view of a computer to software designers
  - Programmer needs architectural details
  - What to do? (Instruction Set, Addressing Modes, Data Types..)

Inside a Computer System

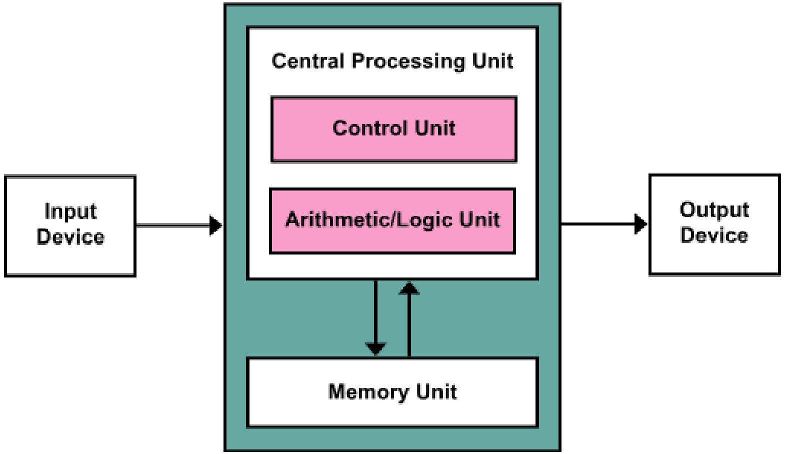




Functional Units of a Computer



Von Neumann Architecture

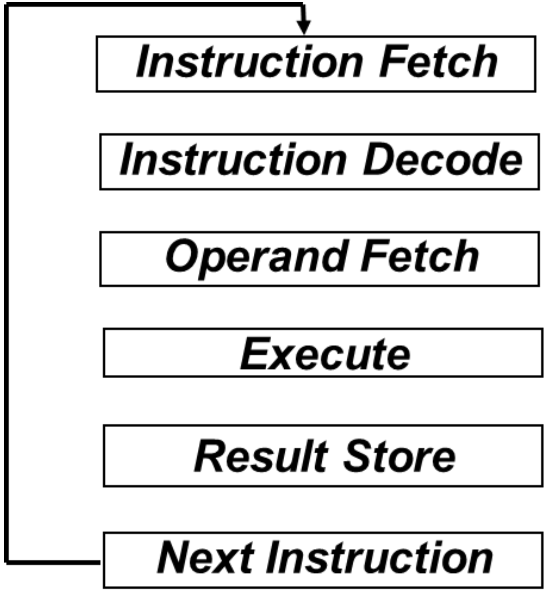


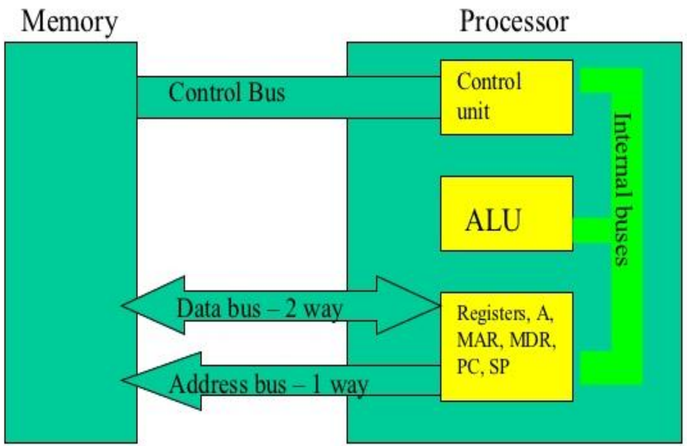
- ❖ The program instructions are stored in memory and are fetched upon requirement to the CPU for executing.
- ❖ Memory is a collection of locations for storing instructions and data.
- ❖ Each location has a unique address which is used to access the location.

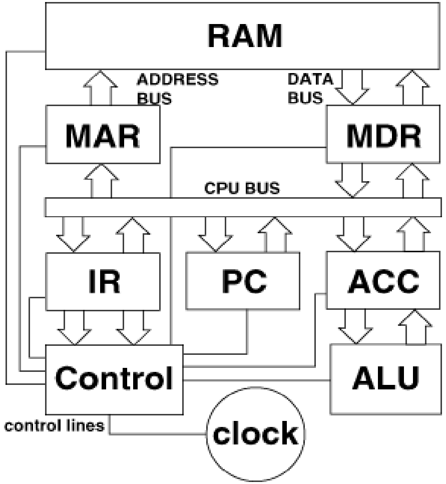
Execution Cycle

Instruction Fetch	Obtain instruction from memory
Instruction Decode	Determine required actions and instruction size
Operand Fetch	Locate and obtain operand data
Execute	Compute result value or status
Result Store	Deposit results in memory for later use
Next Instruction	Determine successor instruction

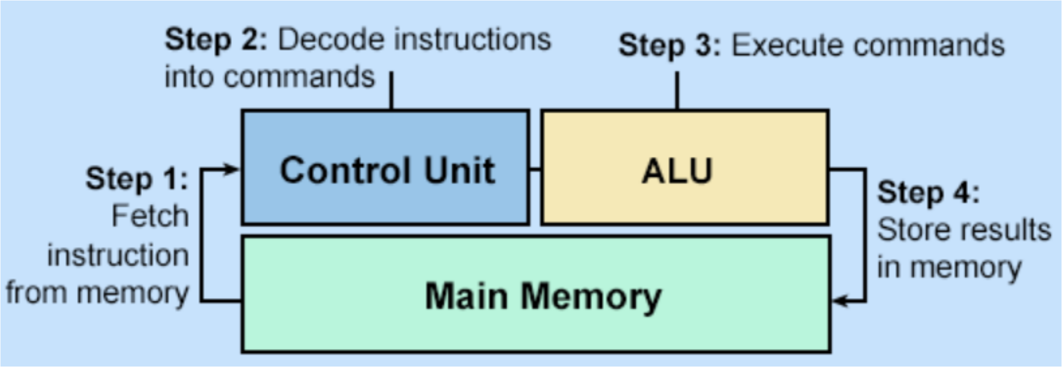
Processor Memory Interaction

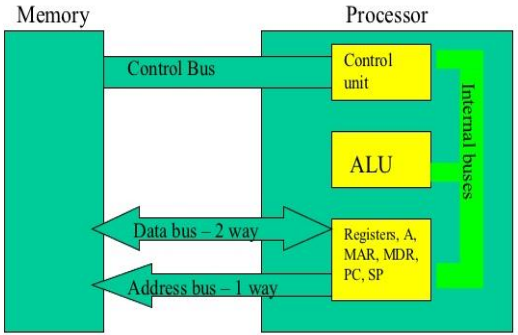


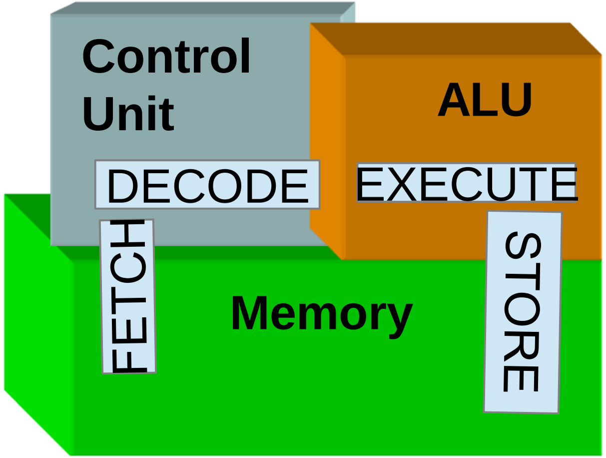




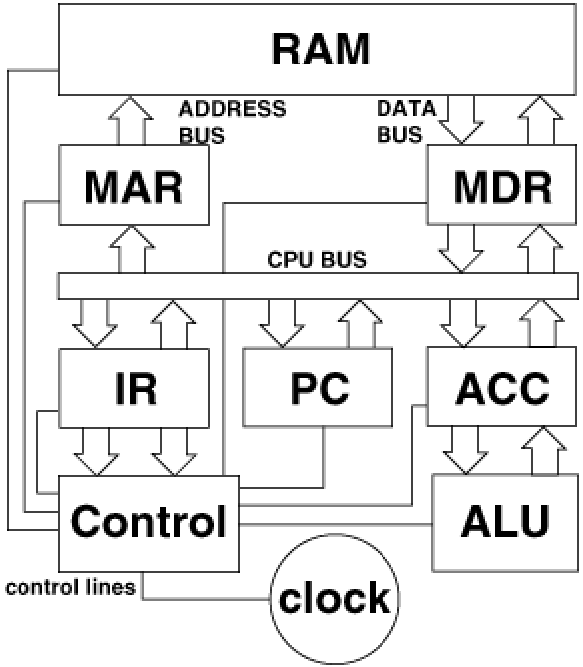
Instruction Execution





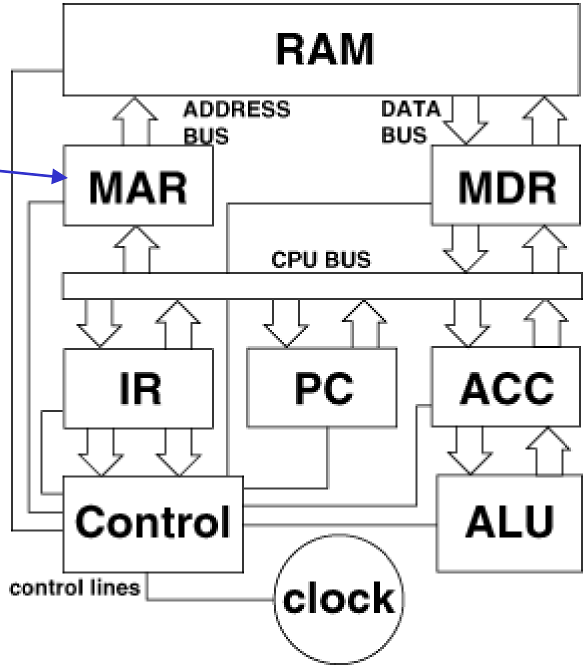


Inside the CPU



Inside the CPU

memory
address
register



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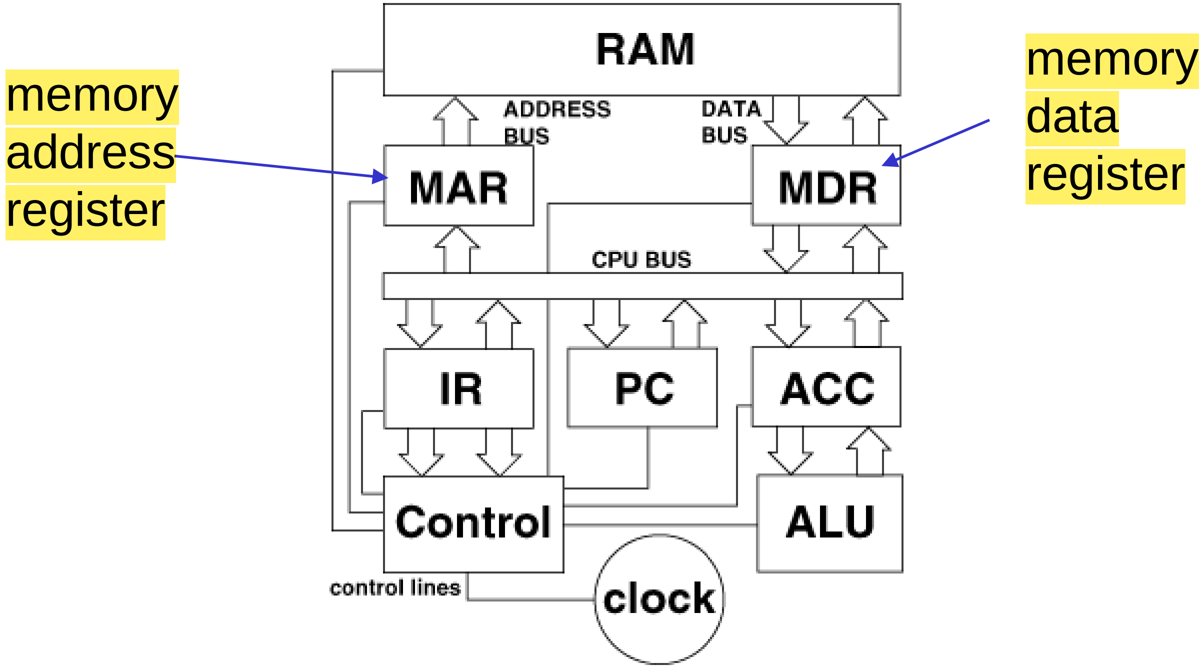

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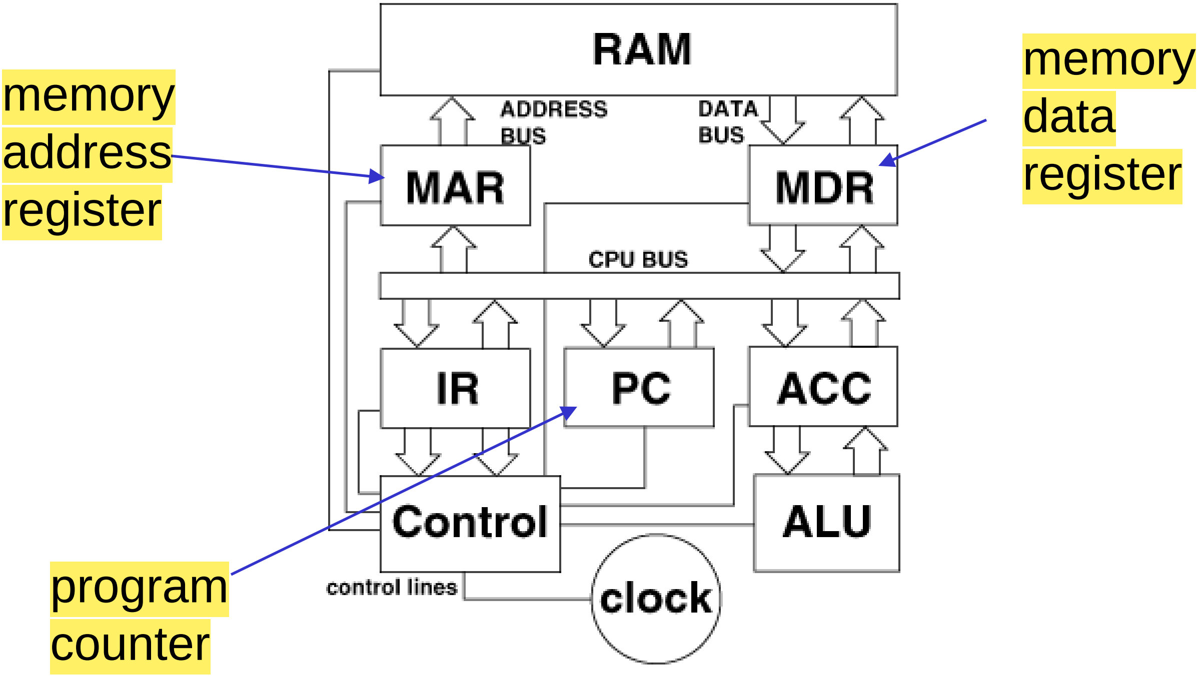

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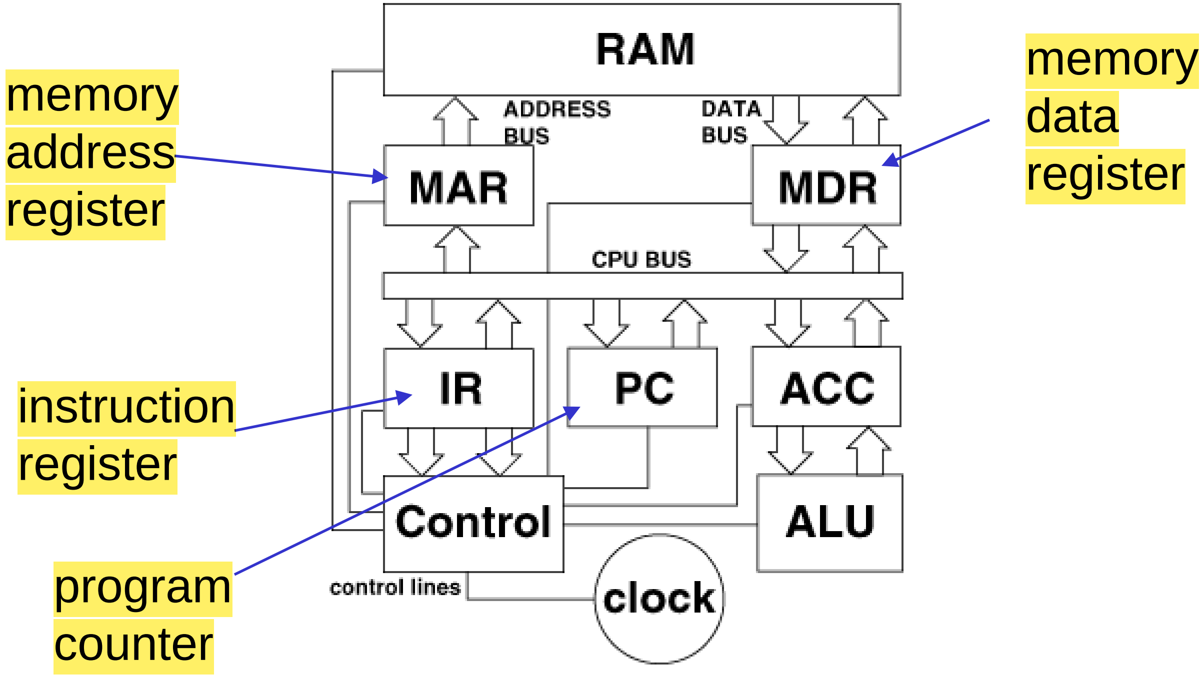

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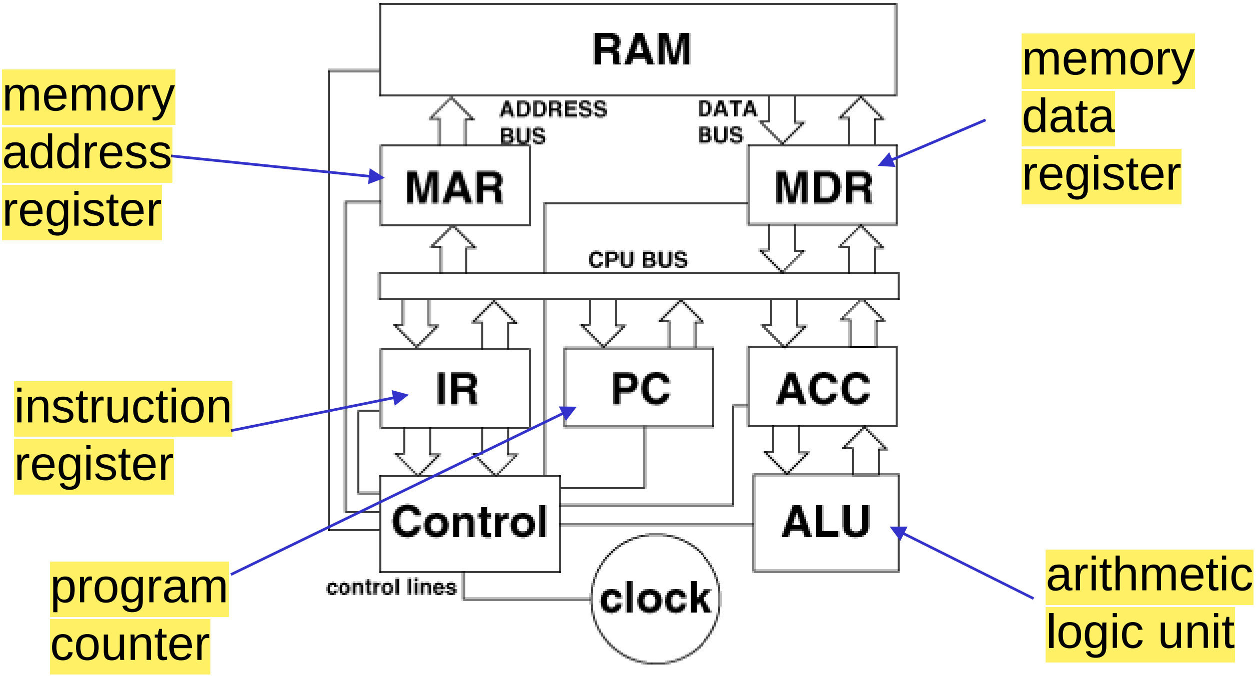

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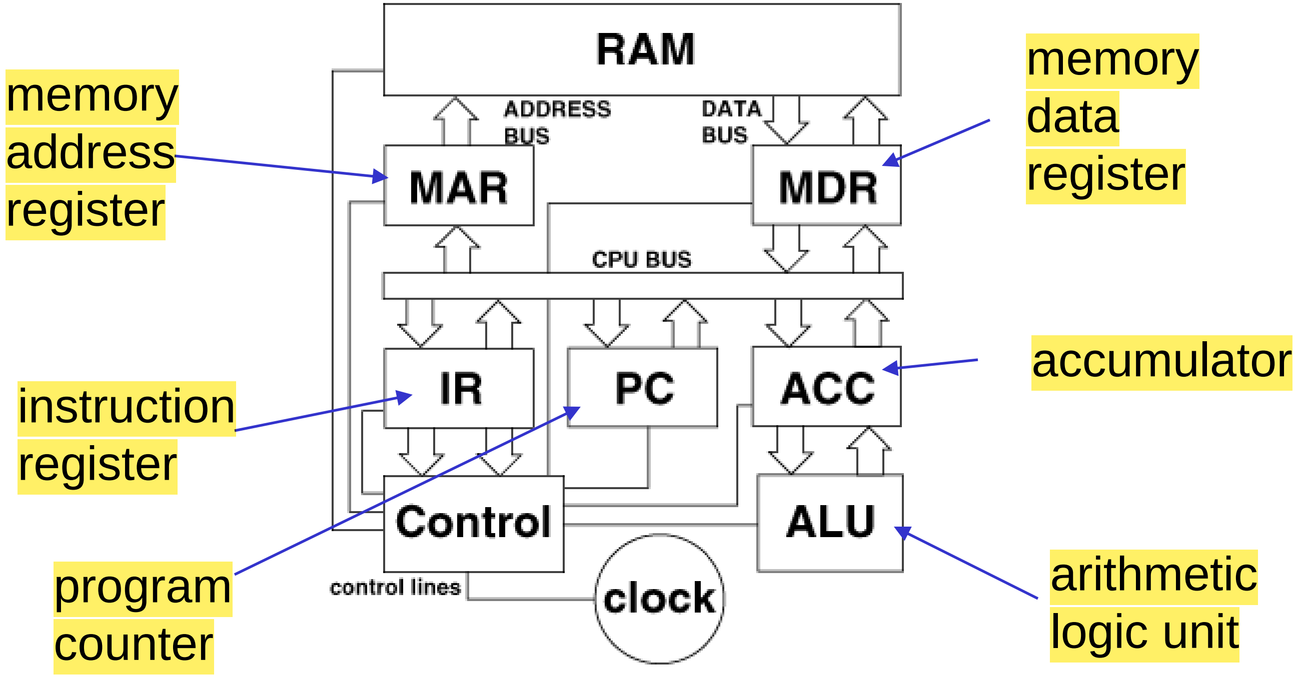
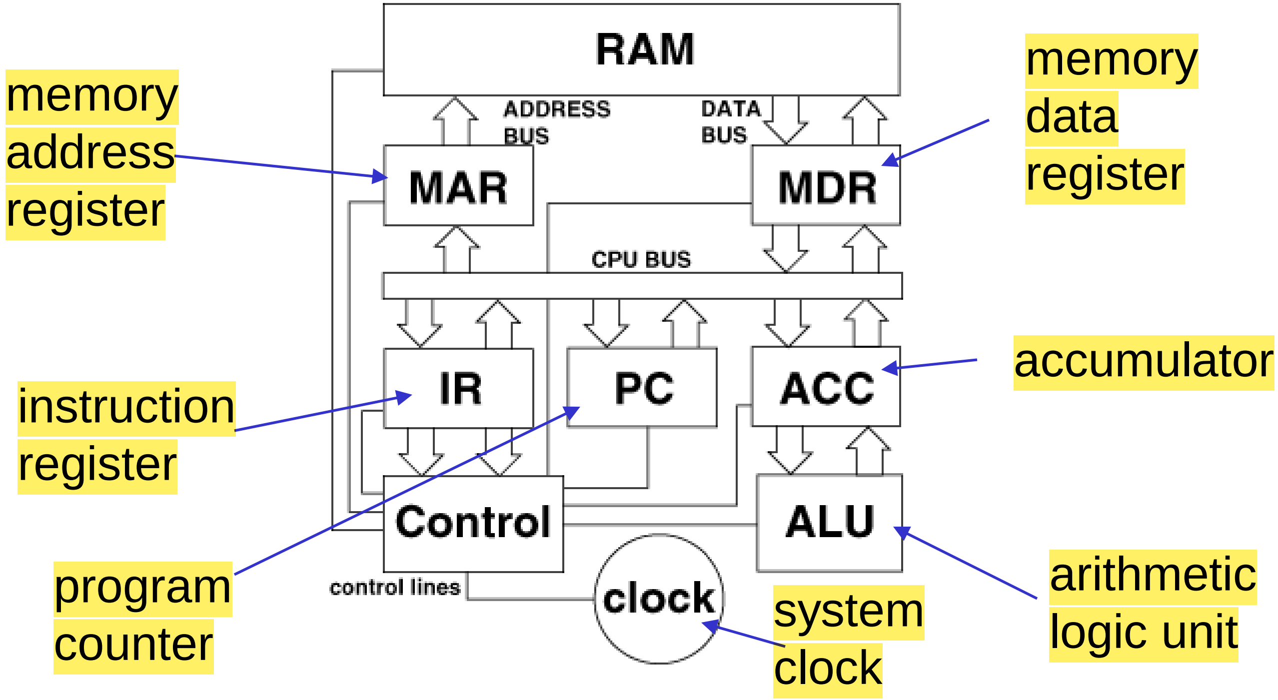
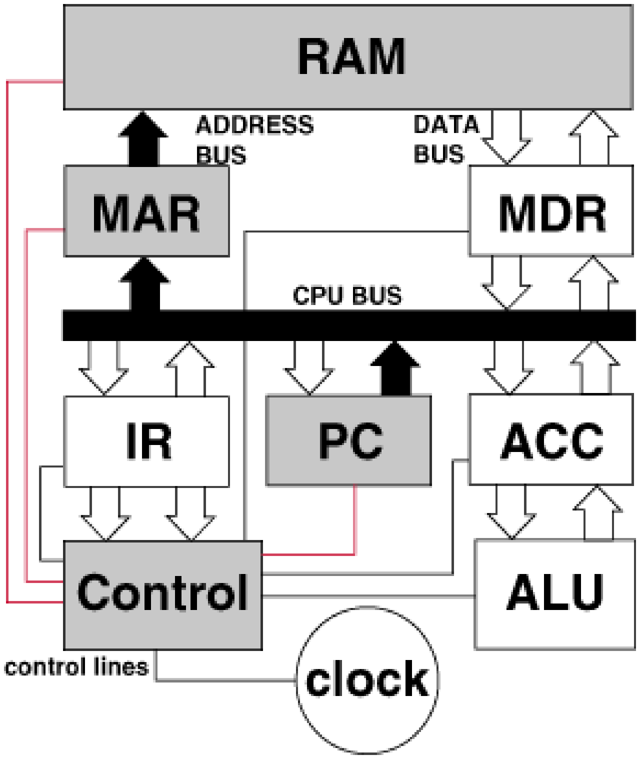


Inside the CPU



FETCH the instruction

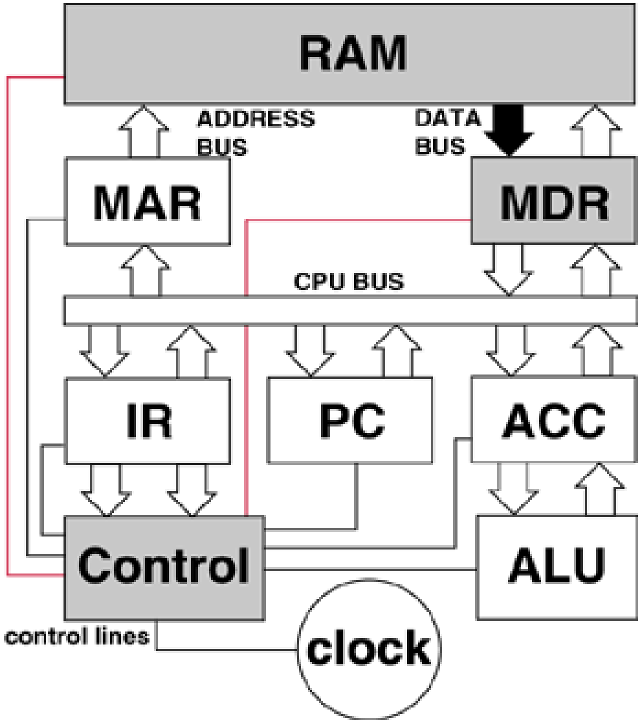


1. Address of the next instruction is transferred from PC to MAR

2. The instruction is located in memory

- Address generated by the CPU comes out to address bus through MAR.
- Address bus is connected to Address decoder.

FETCH the instruction

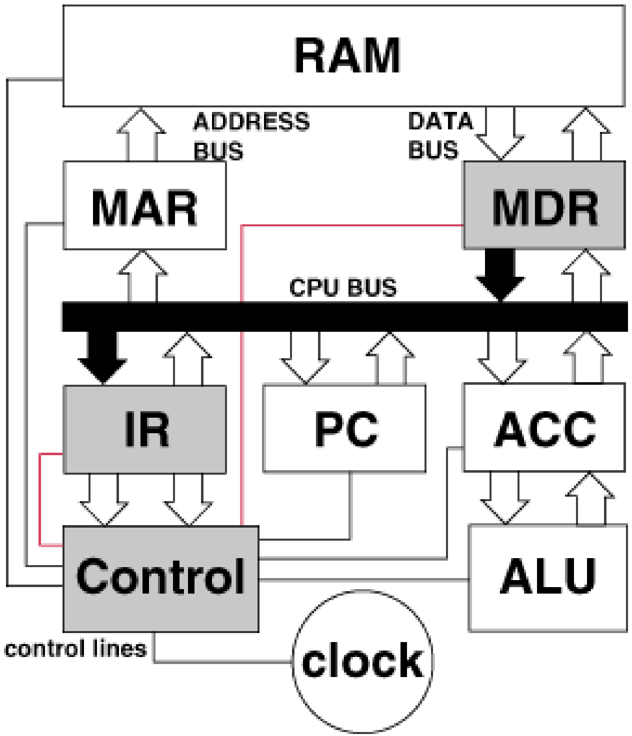


1. Address of the next instruction is transferred from PC to MAR

2. The instruction is located in memory

3. Instruction is copied from to MDR

DECODE the instruction



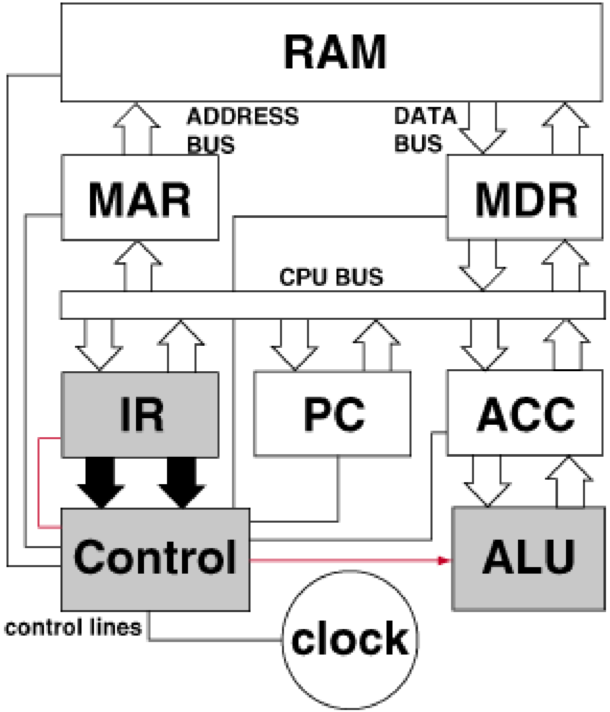
1. Address of the next instruction is transferred from PC to MAR

2. The instruction is located in memory

3. Instruction is copied from to MDR

4. Instruction is moved to IR for decoding

EXECUTE the instruction



1. Address of the next instruction is transferred from PC to MAR

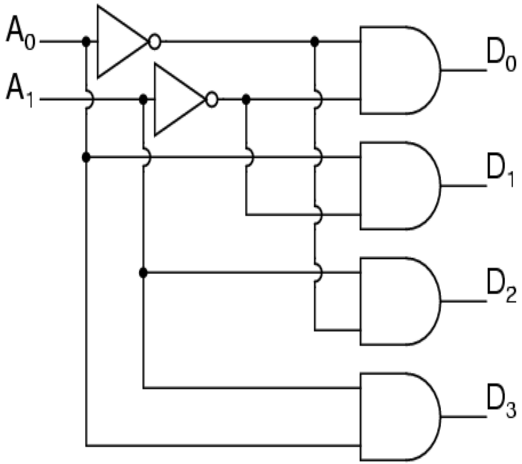
2. The instruction is located in memory

3. Instruction is copied from to MDR

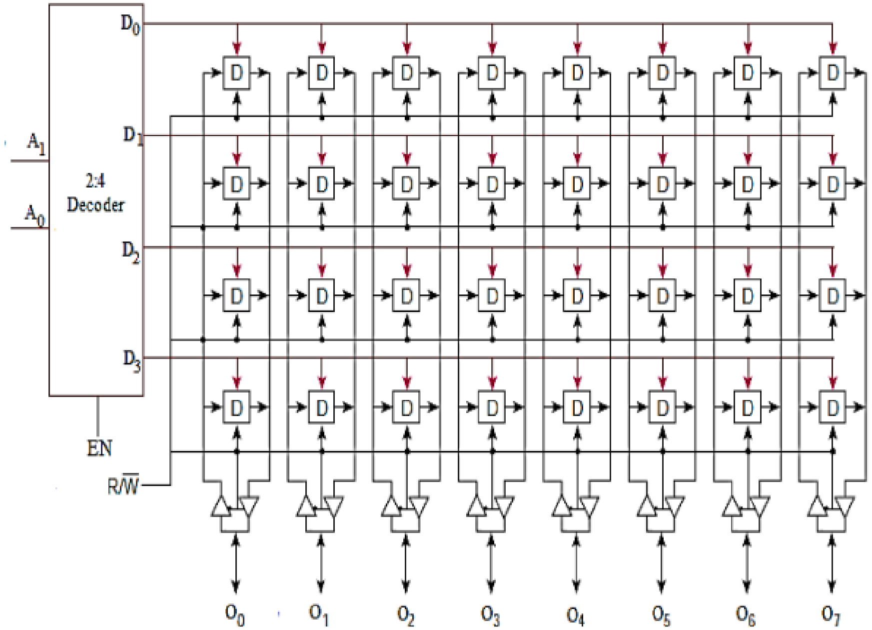
4. Instruction is moved to IR for decoding

5. Control unit sends signals to appropriate devices to cause execution of the instruction

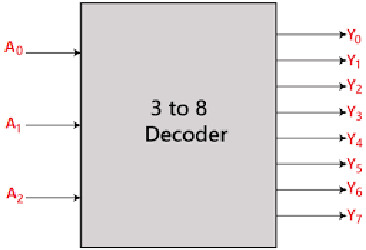
How memory works?

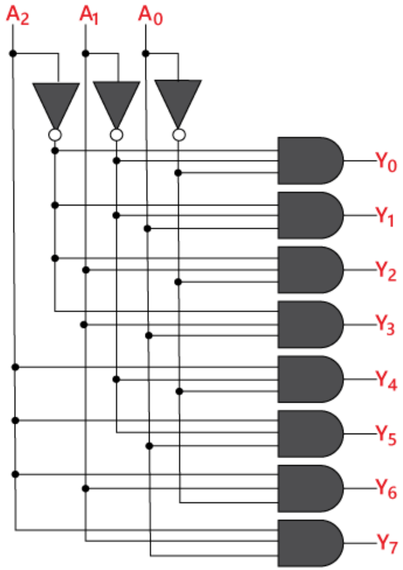


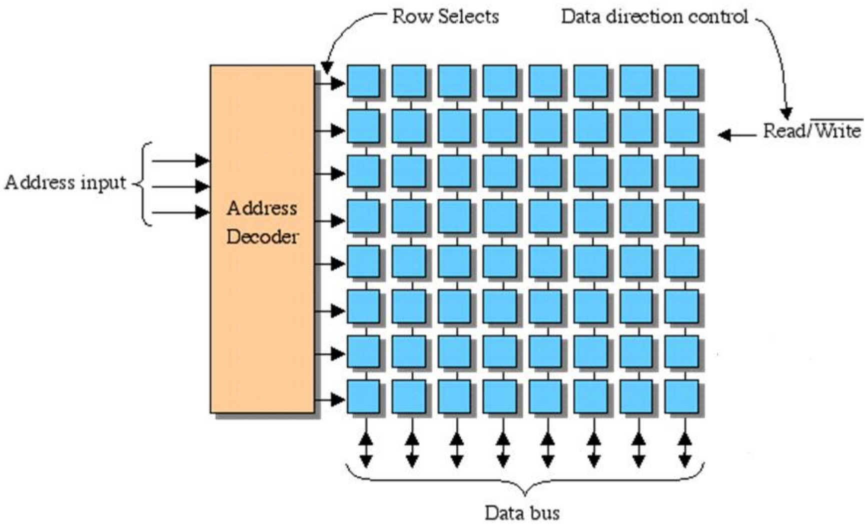
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Address Decoders



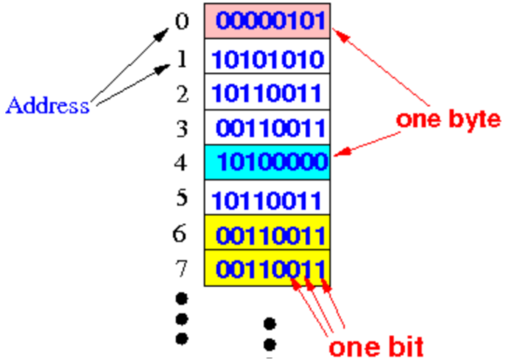


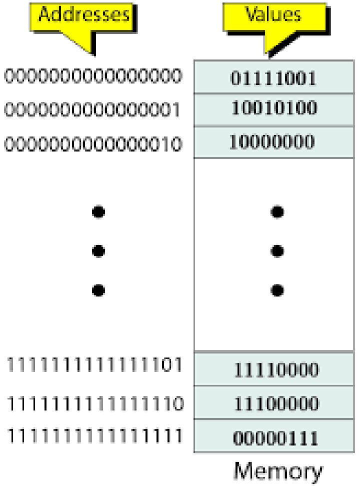


- ❖ One memory location is a collection of 8 bits
- ❖ Each unique address maps to a Byte

Address Bits and Addressability

- ❖ Memory consist of collection of locations
- ❖ 1 location is generally 1 Byte
- ❖ Each location has a unique address
- ❖ 1 bit can uniquely address 2 Bytes [0, 1]
- ❖ 2 bits can uniquely address 4 Bytes
  00, 01, 10, 11 are the 4 unique addresses
- ❖ As number of bits in address increases
  physical address space also increases.





Larger Memory Naming Conventions

8 bits = 1 Byte

1024 Bytes = 1 Kilobyte (KB)

1024 Kilobyte = 1 Megabyte (MB)

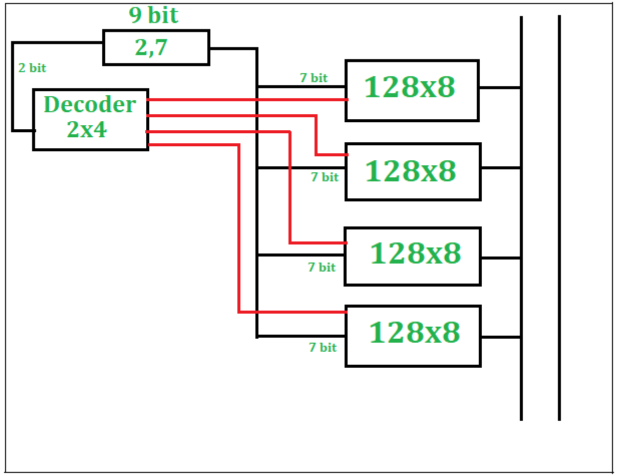
1024 Megabyte = 1 Gigabyte (GB)

1024 Gigabyte = 1 Terabyte (TB)

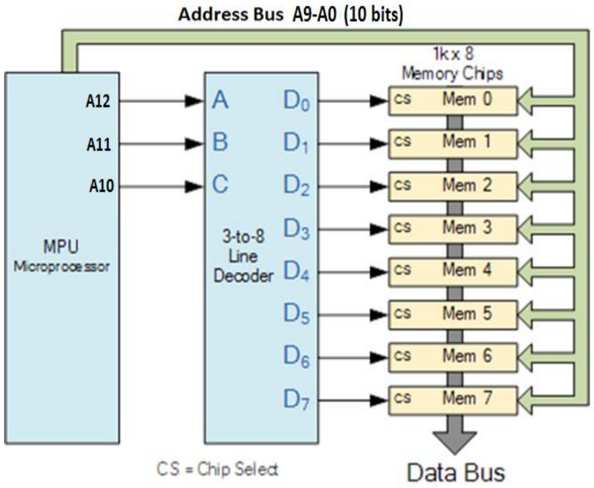
- Kilobyte, megabyte, gigabyte
- How many bytes can be addressed using 8-bit address? : 2^8=256 Bytes
- 64KB→$2^6.2^{10}$ Bytes → 2^{16} Bytes → 16 bit address
- 4GB, 4GB→$2^2.2^{30}$ Bytes → 2^{32} Bytes → 32 bit address

How to make large memory?

- Given a 128 B memory units, how will you create 512 B memory?
- 512 B → 9 bits for address, memory units are 128 B, →7 bit address.



Design of 512 x 8 RAM

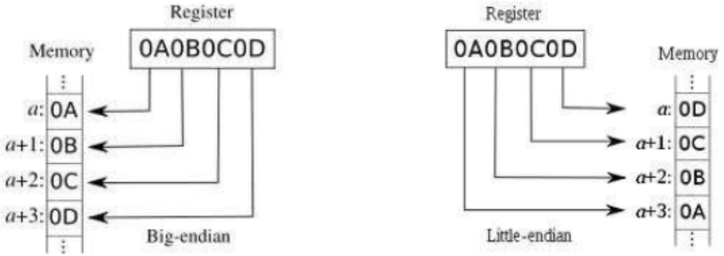


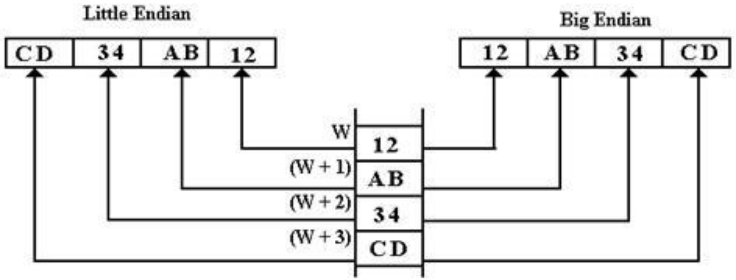
512 B = 4 x 128 B chips

8 KB = 8 x 1 KB chips

Byte Ordering

Big Endian vs. Little Endian





Int i = 450 = $2^8 + 2^7 + 2^6 + 2$ = x000001C2

LSB Little endian

11000010	00000001	00000000	00000000
C2	01	00	00

lower → address higher

MSB Big endian

00000000	00000000	00000001	11000010
00	00	01	C2

Byte Ordering

A 32-bit number 0x23456789 has to be written to memory (collection of bytes) from location 2000 onwards. What is the value stored in location 2001, if the system follows, (a) big endian format and (b) little endian format?

Big Endian

Address	Contents
2000	0x23
2001	0x45
2002	0x67
2003	0x89
2004	

Little Endian

Address	Contents
2000	0x89
2001	0x67
2002	0x45
2003	0x23
2004	

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